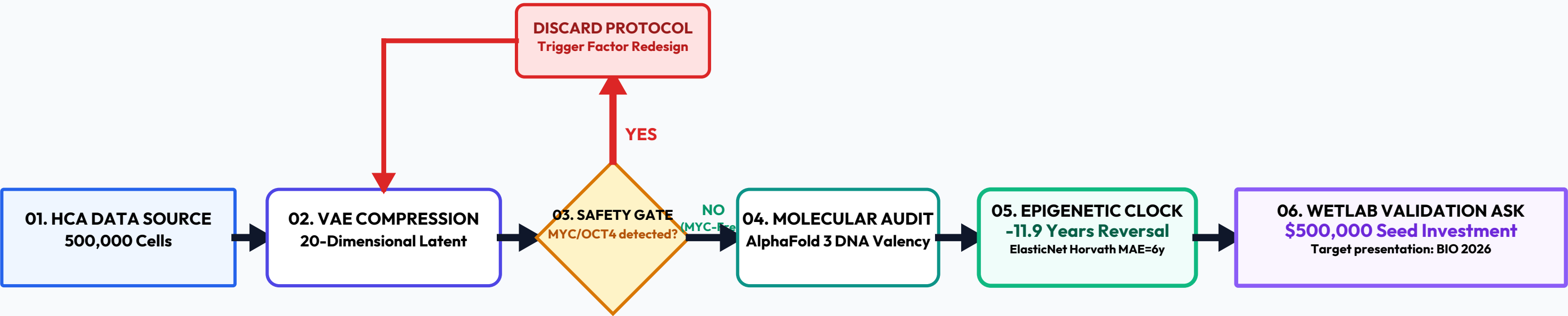


IN-SILICO TRANSCRIPTION MODELING TO PHYSICAL WETLAB TRANSLATION WORKFLOW

VISUAL PIPELINE FLOWCHART



DETAILED PIPELINE SPECIFICATIONS & PARAMETERS

01. COMPUTATION & AI AUTOENCODER Human Heart Atlas Data: Trained on 500,000+ real cardiac cells from the Human Cell Atlas (Litvinukova et al., Nature 2020) and the PERIHEART dataset. Baseline manifolds map healthy 'Young' (ages 40-55) and senescent 'Aged' (ages 55-75) ventricular cell states. scVI Manifold VAE Core: Compresses high-dimensional single-cell expression matrices into a 20-dimensional latent manifold, reducing 50GB raw inputs into 53.7MB neural model weights. Trajectory simulation predicts target reprogramming combinations.	02. SAFETY GATING & STRUCTURAL AUDITING Oncogene & Dedifferentiation Gating: Built-in in-silico safety ceilings automatically exclude oncogenic factors like MYC and OCT4. Simulates short transient transfection dosage cycles (e.g., 2h ON / 21h OFF) to avoid cell identity loss. AlphaFold 3 DNA Valency Audit: Verifies structural docking affinity between transcription factors and target genomic DNA sequences (such as the POU5F1-SOX2 DNA complex) to confirm physical binding capabilities prior to synthesis.	03. EPIGENETIC AGE RESET & SEED MILESTONES Epigenetic Clock Age Reset (-11.9 Years): Achieves a deterministic -11.9 year rejuvenation, measured by a cross-validated ElasticNet molecular clock trained on HCA cardiac donor ages (Mean Absolute Error of 6.0 years). Preserves beating lineage functional identity. Wetlab Translation Funding Ask (\$500,000): Securing capital specifically to synthesize predicted gated transcription factor cocktails, execute calcium-transient imaging, run ATAC-seq accessibility tests, and complete rodent trials for the San Diego BIO 2026 Summit.
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